Digital Twin Anchored Generation: Adaptive Questioning Reveals Directionally Responsive and Resistant Opinion Phenotypes

Ishanu Chattopadhyay

***Abstract*—Digital Twin Anchored Generation (DTAG) selects survey questions using the conditional response structure learned by a Large Science Model (LSM) to direct movement in a pole-relative predictive phenotype. We evaluate this procedure in two response regimes. In the computational experiments, six specified GSS 2024 personas provide the initial states; DTAG chooses each subsequent question and draws its answer from the LSM conditional distribution, without generating persona responses using an LLM. Across 960 twenty-question simulated trajectories, directional policies produced the intended displacement without selecting any of the 27 pole-defining items; 2,880 fixed-set replays assessed the effect of question order. In the prospective human experiment, DTAG again uses LSM-based question selection, but actual participants answer the questions rather than sampling responses from the model. Natural-language answers are mapped to survey variables to update the predictive phenotype. Among 71 completed calibrated human sessions (21 Left, 38 Right, 7 Neutral, 5 Random), 20/21 Left-push participants (95.2%) moved leftward and 21/38 Right-push participants (55.3%) moved rightward. Rightward displacement occurred in 12/29 initially Left-leaning participants (41.4%) and 9/9 initially Right-leaning participants (100%). All 17 initially Left-leaning Right-push nonresponders were identified from their first four adaptive answers, with two additional participants flagged (accuracy 27/29, 93.1%). These results show that adaptive inquiry can produce directional movement in an inferred survey-response phenotype and reveal direction-specific resistance early in the questioning sequence.**

***Index Terms*—Predictive phenotypes, large science models, conditional response modeling, digital twins, survey methodology, opinion dynamics, adaptive questioning, generative simulation.**

I. Introduction

Survey questions are usually treated as measurements, but each answer also changes predictions about unanswered items. We ask whether choosing *which questions to pose* can steer a respondent's inferred ideological position. If so, adaptive surveys do more than collect information: their selection policies shape the modeled opinion profile.

Large Science Models (LSMs) provide a new framework for generative digital twins of multivariate systems. They learn interdependent conditional distributions, infer unobserved variables from partial evidence, and compare states through their predictive consequences. Related digital-twin applications include infant microbiome development [1] and influenza evolution through Emergenet [2]. DTAG applies an LSM to survey answers, defining a *predictive phenotype* over unasked items and its position relative to opposing reference opinion profiles. It selects questions to maximize expected signed displacement while respondents supply the answers.

We test DTAG first in 960 native GSS 2024 simulations seeded by six stated personas, with answers sampled entirely from the LSM, followed by 2,880 fixed-set LSM replays. We then replace the simulated respondents with human participants: the adaptive questions remain LSM-directed, but the answers come from 71 Prolific participants and update the same conditional-survey phenotype through native-variable mappings. The human trajectories also reveal baseline-dependent Right-push resistance detectable from the first four responses.

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II. Related Work

Attitude dependencies need not align on one ideological axis [3], [4]. Partisan sorting and social influence shape these dependencies [5], [6]; cognitive consistency, wording, and order can affect reported attitudes [7]–[9]. This motivates separating information revealed by a question from any change induced by asking it.

Opinion-dynamics models often use low-dimensional transitions [10]–[12], while experiments on exposure and contact report heterogeneous effects [13]–[15]. Generative agents and prompted survey personas can produce plausible responses [16], [17]. DTAG instead combines learned high-dimensional survey conditionals with an explicit question-selection objective.

III. Large Science Models and the Predictive Phenotype

A. Conditional dependency model

Let $X = (X_1, \ldots, X_n)$ be categorical survey responses, with alphabets $\mathcal{A}_i$ and survey context c. An LSM comprises conditionals

$$\Phi_c = \{\phi_{i,c}(a \mid x_{-i}) : i = 1, \ldots, n\}, \tag{1}$$

estimated by conditional inference trees [1], [18], [19]. The learned conditionals supply both the question-selection objective and the updated predictive phenotype.

A respondent state $s = \{X_j = a_j : j \in S\}$ holds observed responses. Unanchored variables remain missing in $\widetilde{x}_{-v}(s)$; each native tree averages over its missing branches. For unasked $v \notin S$,

$$p_\Phi(a \mid s, v) = \phi_{v,c}(a \mid \widetilde{x}_{-v}(s)). \tag{2}$$

The vector

$$\mathcal{B}_\Phi(s) = \big(p_\Phi(\cdot \mid s, v)\big)_{v \notin S} \tag{3}$$

is the *predictive phenotype*: conditional distributions and uncertainty for unasked responses.

B. Geometry as predictive equivalence

Conditional predictions induce a distance between complete or partial states x, y:

$$\theta_\Phi(x,y) = \frac{1}{n}\sum_{i=1}^{n}\sqrt{D_{\mathrm{JS}}(\phi_{i,c}(\cdot \mid x_{-i}), \phi_{i,c}(\cdot \mid y_{-i}))}. \quad (4)$$

Here D_{JS} denotes Jensen–Shannon divergence. The resulting predictive pseudometric distinguishes states by their conditional consequences.

Prior LSM reconstruction reduced mean distance distortion by 54.9% (GSS) and 60.9% (Eurobarometer) with 80% of responses masked [19].

For GSS models with registered left and right reference states s_L, s_R, DTAG reports

$$I_\Phi(s) = \frac{\theta_\Phi(s_L, s) - \theta_\Phi(s_R, s)}{\theta_\Phi(s_L, s_R)}. \quad (5)$$

The reference poles specify opposing responses on 27 GSS 2024 items covering policy, civil liberties, spending, and religion (Table I) [19]. Other coordinates are unspecified; I_Φ measures position in the predictive geometry induced by the LSM.

IV. Adaptive Question Selection

DTAG separates *question selection* from the *source of the answer*. The LSM conditionals determine the expected geometric effect of each candidate question. In the native simulations, a fixed initial persona anchors the state, the optimizer chooses survey item v_t, and the answer is drawn directly from the LSM:

$$A_t \sim p_\Phi(\cdot \mid s_t, v_t), \quad (6)$$

$$s_{t+1} = s_t \cup \{v_t = A_t\}, \quad (7)$$

$$\mathcal{B}_{t+1} = \mathcal{B}_\Phi(s_{t+1}). \quad (8)$$

These native trajectories use no LLM to generate respondents' answers. In the human experiment, the LSM still supplies the directional selection criterion, but the selected question is presented to a participant, whose *observed answer replaces the LSM draw*. A language-model interface renders questions and interprets free-text responses as one or more native survey-variable assignments; these accepted assignments update the conditional state. Thus the displayed single-item update describes the native simulation exactly, while the human implementation can incorporate multiple assignments from one answer. In neither regime does the question-selection algorithm prescribe the answer.

For candidate item v and answer a, define

$$\Delta_v(a; s) = I_\Phi(s \cup \{v = a\}) - I_\Phi(s), \quad (9)$$

$$\mu_v(s) = \sum_{a \in \mathcal{A}_v} p_\Phi(a \mid s, v)\Delta_v(a; s), \quad (10)$$

$$\sigma_v^2(s) = \sum_a p_\Phi(a \mid s, v)(\Delta_v(a; s) - \mu_v(s))^2. \quad (11)$$

Right- and left-seeking policies maximize and minimize $\mu_v(s)$; neutral minimizes $|\mu_v(s)|$, and random samples an eligible item. By construction,

$$\mathbb{E}[I_\Phi(s_{t+1}) - I_\Phi(s_t) \mid s_t, v_t] = \mu_{v_t}(s_t).$$

Accumulated displacement follows successive selected items and their realized answers, which are LSM draws in simulation and participant responses in the human experiment.

The effect on an unasked item j after observing (v, a) is

$$R_j(s; v, a) = D_{\mathrm{JS}}\big(p_\Phi(\cdot \mid s, j), p_\Phi(\cdot \mid s \cup \{v = a\}, j)\big). \quad (12)$$

A weighted sum $\mathcal{P}_\Phi(s; v, a) = \sum_{j \notin S \cup \{v\}} w_j R_j(s; v, a)$ measures redistribution across the remaining items. Unlike text retrieval [20], DTAG maintains an explicit conditional response state. The public implementation includes a web application (https://github.com/zeroknowledgediscovery/DTAG/tree/main/webapp).

V. Refinement and Possible Response Effects

Under *phenotypic refinement*, successive answers restrict predictions for unasked items. For held-out responses Y, the expected information from asking v is

$$\mathcal{G}(v \mid s) = H(Y \mid s) - \mathbb{E}_{A_v|s}[H(Y \mid s, v = A_v)]. \quad (13)$$

Under a coherent joint model this is conditional mutual information, distinct from directional displacement $\mu_v(s)$. Held-out response prediction evaluates refinement directly.

A. Possible response effects.

Questioning a person may change later reports. A response–transition kernel separates an observed answer from a possible latent-state update:

$$K_q(a, z' \mid z) = O_q(a \mid z)\, T_q(z' \mid z, a). \quad (14)$$

Passive observation has $T_q(z' \mid z, a) = \delta_z(z')$; shifts in salience or belief can induce nontrivial transitions. This transition model distinguishes the measurement process from possible response-dependent state transitions.

B. Order dependence.

For fixed realized answers, the terminal index is order-independent; sequentially generated answers need not be. A coherent joint distribution satisfies

$$P(a \mid s, v)P(b \mid s, v = a, w) = P(b \mid s, w)P(a \mid s, w = b, v). \quad (15)$$

Separately fitted LSM conditionals may violate this identity [21]. Fixed-set replays evaluate its consequences for native response sequences.

VI. Native GSS 2024 Computational Experiments

A. Adaptive questioning from pinned persona states

We initialized the LSM with six explicitly specified GSS 2024 persona states (Table II) and generated 40 independent trajectories per persona and question-selection policy (right, left, neutral, random). At each of twenty steps the directional policy selected a survey item from the LSM's predicted effects, then a response was *sampled from that item's LSM conditional distribution* and incorporated into the state. This is an LSM-driven simulation, not an LLM-generated persona dialogue. *All 27 pole-defining items*

TABLE I

GSS 2024 reference poles. TWENTY-SEVEN PAIRED RESPONSES DEFINE s_L, s_R, WITH $I_\Phi = -1, +1$; ALL OTHER COORDINATES ARE UNSPECIFIED.

Item	Survey topic	Left	Right
abany	Abortion on request	yes	no
abdefect	Abortion: fetal defect	yes	no
abnomore	Abortion: no more children	yes	no
abpoor	Abortion: low income	yes	no
abrape	Abortion after rape	yes	no
absingle	Abortion: unmarried	yes	no
bible	Bible interpretation	ancient book	inspired word
colcom	Communist as college teacher	not fired	yes, fired
confed	Trust in federal government	a great deal	hardly any
conlabor	Trust in organized labor	a great deal	hardly any
grass	Marijuana legalization	should be legal	should not be legal
gunlaw	Permit to purchase gun	favor	oppose
libcom	Communist books in library	not remove	remove
libmslm	Muslim cleric books in library	not remove	remove

Item	Survey topic	Left	Right
natarms	Military spending	too much	about right
natenvir	Environmental spending	too little	about right
natfare	Welfare spending	too little	too much
natsoc	Social Security spending	too little	about right
owngun	Guns in household	no	yes
pillok	Contraception access	strongly agree	disagree
polabuse	Police strike insulting citizen	yes	no
pray	Frequency of prayer	never	several times a day
prayer	Bible prayer in public schools	approve	disapprove
religen	Personal religiosity	no religion	strong
rowngun	Respondent owns a gun	no	yes
shotgun	Owns a shotgun	no	yes
spkcom	Communist public speech	yes, allowed to speak	not allowed

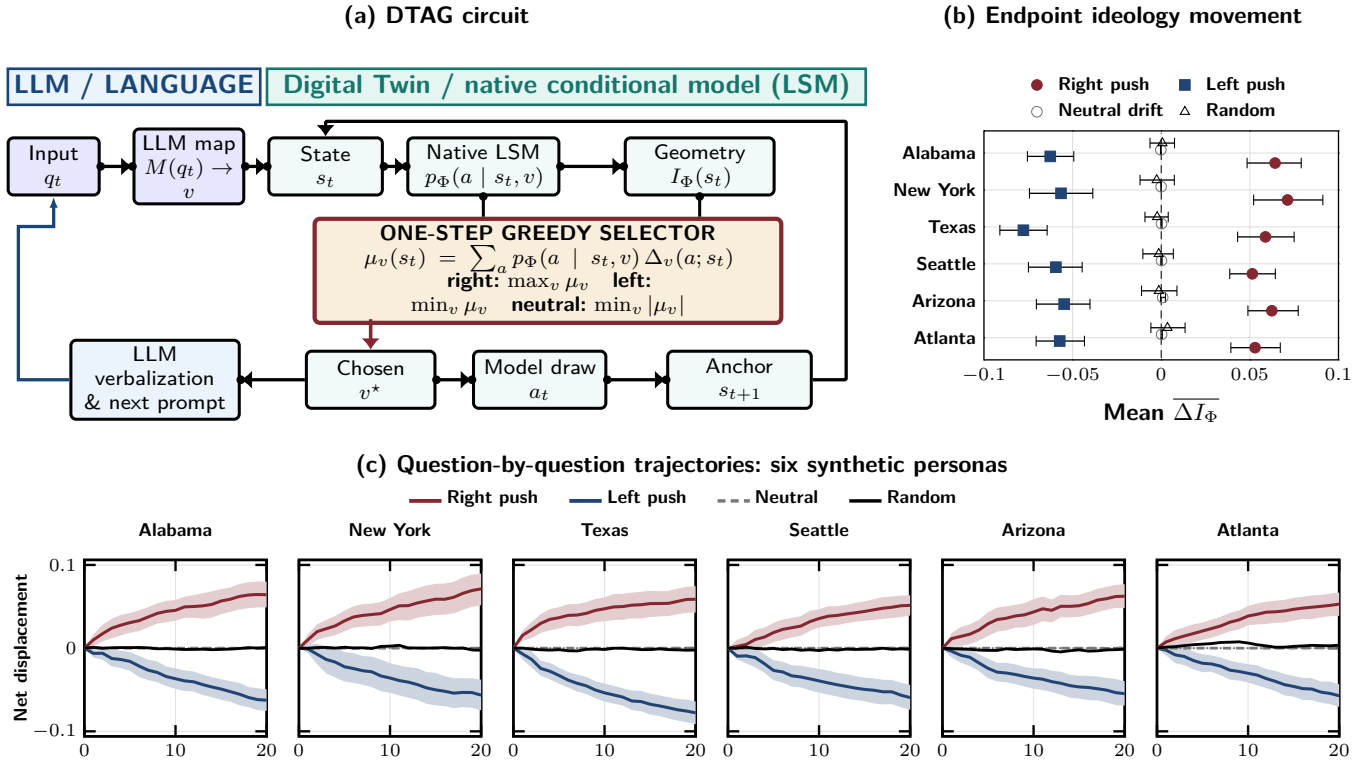


Fig. 1. **DTAG architecture and native adaptive results.** (a) LSM-anchored responses and directional question selection; language mapping is a separate interface. (b) Mean endpoint displacement across six personas with 95% bootstrap intervals. (c) Mean 20-question trajectories over 40 repetitions per persona and policy, with pointwise 95% intervals (right: red; left: blue; neutral: dashed gray; random: black).

were excluded from question selection (Table I). The 960 trajectories therefore comprise 19,200 responses to non-pole items, with pinned starting states and independently sampled answers.

Across all six personas, right- and left-seeking policies moved the index in their intended directions (Fig. 1). As pole-defining questions were excluded, the movement results from conditioning on other survey items, not directly asking those used to define the reference index. Neutral and random policies provide comparisons.

B. Question selection as observed paths

Question identities are summarized in Table III. The 18 displayed heatmaps contain 161 distinct items across right, left, and neutral selection (Fig. 2). A shared categorical ordering approximately minimizes

$$\mathcal{C}(\pi) = \sum_{p,u,t} \sum_{r<r'} \left|\pi(v_{purt}) - \pi(v_{pur't})\right|, \qquad (16)$$

Here u indexes policies. Heatmap colors identify items, not ideological position; 67 further items occurred only in random selection.

TABLE II
PINNED GSS 2024 SYNTHETIC PERSONAS AND BASELINE POLE-RELATIVE INDEX.

Persona	**Pinned description**	**Initial $I_\Phi(s_0)$**
Alabama	45-year-old rural Alabama white man; married, two children; full-time farmer and veteran; high school; conservative, strong Republican; Protestant, weekly attendance; daily newspaper.	0.075706
New York	22-year-old New York City white woman; unmarried, no children; full-time retail worker, nonveteran; bachelor's degree; extremely liberal, strong Democrat; nonreligious; daily newspaper.	−0.163807
Texas	61-year-old small-town Texas white woman; married, three children; homemaker, high school; extremely conservative, strong Republican; Protestant, attends services several times weekly; daily newspaper.	0.045379
Seattle	34-year-old Seattle white man; unmarried, no children; employed full time, graduate degree; liberal, weak Democrat; nonreligious; newspaper several times weekly.	−0.119639
Arizona	39-year-old Mexican American man in an Arizona city; married, three children; employed full time, high school; conservative, weak Republican; Catholic, nearly weekly attendance.	−0.020437
Atlanta	47-year-old Atlanta Black woman; divorced, two children; employed full time, bachelor's degree; liberal, strong Democrat; Protestant, weekly attendance; daily newspaper.	−0.030807

Positive values lie nearer the right reference pole; negative values nearer the left.

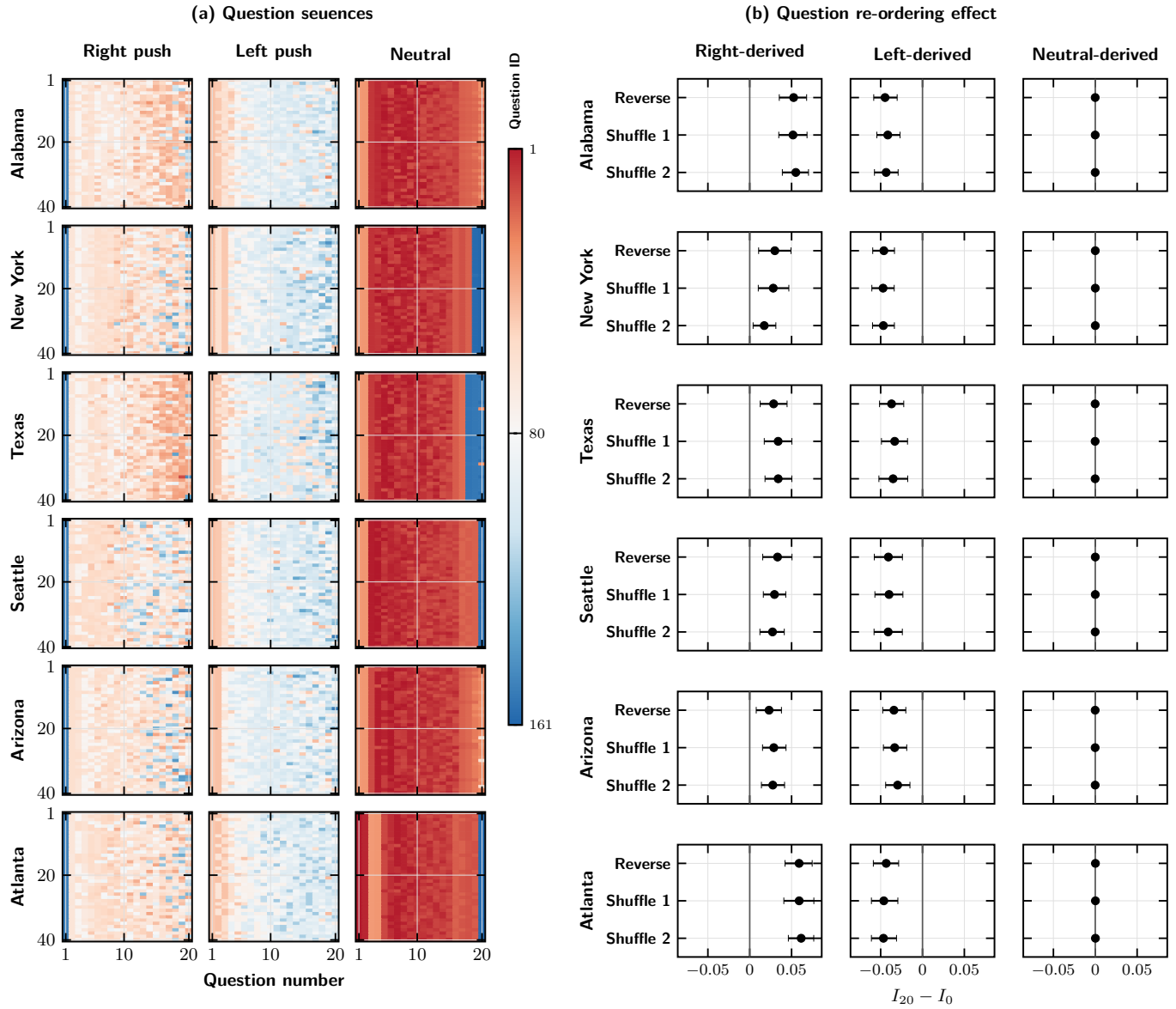


Fig. 2. **Selected questions and fixed-set replays.** Left: question-ID heatmaps for six personas and three selection policies (40 runs each); color denotes item identity. Right: mean index displacement and 95% bootstrap intervals for 20-item sets replayed in original, reversed, and two shuffled orders. Right- and left-derived sets retain their direction; neutral sets remain near zero.

TABLE III
Question selection in the native GSS 2024 experiment.

Selection policy	Distinct	Selections	Pole overlap
Right-seeking	100	4,800	0
Left-seeking	75	4,800	0
Neutral	43	4,800	0
Three-policy union	161	14,400	0
All four policies	228	19,200	0

The random policy contributed 4,800 selections; 67 of the 228 distinct items occurred only under random selection.

C. Fixed-set replay: robustness to question ordering

For each persona and right/left/neutral policy, we froze the 20-item sequence from adaptive replicate zero, giving 18 fixed sets. Each set was replayed in original, reversed, and two shuffled orders, drawing 40 fresh response sequences per order directly from the LSM (2,880 runs). Matched question-specific random variates reduced contrast variance.

The direction persisted across replays (Fig. 2): right-derived sets yielded mean displacement +0.0381 (original) and +0.0377 (reversed); left-derived sets yielded −0.0399 and −0.0410. In both orders, all six persona-specific 95% bootstrap intervals excluded zero in the prescribed direction. Shuffles followed the same pattern; neutral sets remained near zero.

All 54 pointwise bootstrap intervals for paired mean between-order contrasts included zero; no systematic mean order effect was detected. Selected question sets therefore retained directional displacement without continuous reoptimization.

VII. Human Response Study

The human experiment applies the LSM-based directional question-selection procedure to actual participants instead of simulated respondents. Questions are selected using the learned conditionals, but the participant's own answer—interpreted as native survey assignments—determines the next state. The 71 completed calibrated sessions and the early Right-push resistance result are reported below.

A. Protocol and outcome

We recruited participants through Prolific for the prospective DTAG human-response experiment. Each calibrated session comprised six baseline questions, twenty adaptive questions, and five interleaved attention checks. DTAG selected the adaptive questions from the LSM's expected directional effects, as in the native simulations. In contrast to those simulations, responses were supplied by human participants rather than drawn from the LSM conditional distributions. Participants answered in natural language; an LLM-based interface presented the questions and mapped each free-text response into one or more accepted native survey-variable assignments, which updated the LSM state. The LLM did not generate the participants' answers. Four question-selection policies were used: Left, Right, Neutral, and Random. The original study randomized policies; subsequent cohorts recruited participants under fixed Left or Right assignments.

Let s_0 be the state after the six baseline questions and s_t the state following adaptive answer t. The index trajectory and baseline-relative displacement are

$$\Delta I_\Phi(t) = I_\Phi(s_t) - I_\Phi(s_0), \qquad t = 0, \ldots, 20. \tag{17}$$

A directional success at the end of the adaptive sequence is $\Delta I_\Phi(20) < 0$ for Left and $\Delta I_\Phi(20) > 0$ for Right. The resulting trajectories follow the successive human-supplied answers in the LSM predictive geometry.

B. Completed sessions and directional response

The calibrated study contains 71 participants who completed the 20 adaptive questions: 25 from the original four-policy experiment, 15 from the fixed-Left follow-up, and 31 from the fixed-Right follow-up. The assigned-policy totals are Left ($n = 21$), Right ($n = 38$), Neutral ($n = 7$), and Random ($n = 5$). The separately recruited zero-calibration Right cohort is not combined with these participants, since it does not have an individually measured six-question baseline.

The archived results show leftward displacement in 20/21 Left-push participants (95.2%) and rightward displacement in 21/38 Right-push participants (55.3%). Baseline position strongly distinguishes the observed Right-push response: 12/29 initially Left-leaning participants (41.4%) moved right, compared with 9/9 initially Right-leaning participants (100%). For Left push, 16/17 initially Left-leaning participants (94.1%) and 4/4 initially Right-leaning participants (100%) moved left.

The mean endpoint displacement was −0.0989 for Left push (95% participant-bootstrap confidence interval $[-0.1200, -0.0741]$) and +0.0225 for Right push ($[+0.0006, +0.0452]$). Neutral and Random showed smaller mean displacements (−0.0054 and −0.0045, respectively).

C. Early identification of Right-push resistance

Right-directed questioning produced a weaker response among initially Left-leaning participants. We therefore examined whether resistance to rightward displacement could be identified from the first four adaptive responses. This analysis was restricted to the 29 Right-push participants whose baseline inferred ideology index was negative. A participant was flagged as resistant if their inferred index had not increased relative to their own baseline after four questions. We evaluated that early classification against whether the index had moved rightward from baseline at the end of the full 20-question sequence.

Seventeen of these 29 participants showed no net rightward displacement at the end of the study. All 17 had been identified by the four-question criterion. The criterion also flagged two participants who ultimately moved rightward. It therefore identified all observed resistant participants (17/17, 100% sensitivity), with 89.5% precision (17/19), 83.3% specificity (10/12), and 93.1% overall classification accuracy (27/29).

TABLE IV

DIRECTIONAL RESPONSE AFTER THE 20 ADAPTIVE QUESTIONS, STRATIFIED BY THE SIGN OF THE BASELINE INFERRED IDEOLOGY INDEX. SUCCESS DENOTES NET DISPLACEMENT IN THE ASSIGNED DIRECTION.

Initial inferred phenotype	Left push	Right push
Initially Left	16/17 (94.1%)	12/29 (41.4%)
Initially Right	4/4 (100%)	9/9 (100%)
Overall	20/21 (95.2%)	21/38 (55.3%)

TABLE V

HUMAN-STUDY RESULTS BY ASSIGNED POLICY. ENDPOINT DISPLACEMENT IS MEASURED FROM EACH PARTICIPANT'S SIX-QUESTION BASELINE; INTERVALS ARE 95% PARTICIPANT-BOOTSTRAP PERCENTILE INTERVALS (5,000 REPLICATES).

Policy	n	Baseline $I_\Phi < 0$	Baseline $I_\Phi > 0$	Mean $\Delta I_\Phi(20)$ (95% CI)	Directional success
Left	21	17	4	-0.0989 $[-0.1200, -0.0741]$	20/21 (95.2%)
Right	38	29	9	$+0.0225$ $[+0.0006, +0.0452]$	21/38 (55.3%)
Neutral	7	6	1	-0.0054 $[-0.0153, +0.0032]$	—
Random	5	2	3	-0.0045 $[-0.0512, +0.0413]$	—

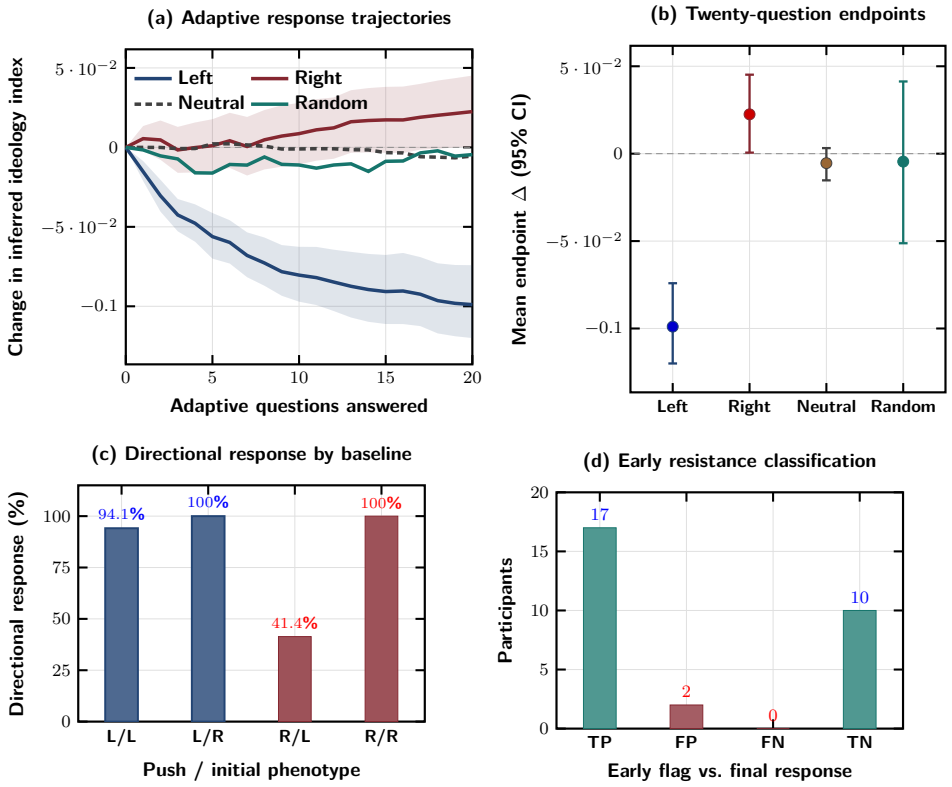


Fig. 3. **Updated human-study trajectories and endpoints (71 calibrated complete sessions).** (a) Mean baseline-relative change after each adaptive question in Left, Right, Neutral, and Random policies; the colored 95% participant-bootstrap confidence bands are shown for Left and Right. (b) Mean displacement after twenty adaptive questions, with 95% participant-bootstrap intervals from complete human-response trajectories. (c) Proportion moving in the assigned direction by initial phenotype sign and policy: L/L = 16/17, L/R = 4/4, R/L = 12/29, and R/R = 9/9, where the first symbol denotes policy and the second denotes baseline sign. (d) Classification from the first four questions of baseline-relative Right-push nonresponse at the end of the study among initially Left-leaning participants: 17 true positives (TP), 2 false positives (FP), no false negatives (FN), and 10 true negatives (TN).

D. Interpretation

The human-response trajectories exhibit a marked directional asymmetry. Left-directed questioning moved 20/21 inferred phenotypes leftward; Right-directed questioning moved all nine initially Right-leaning participants but only 12/29 initially Left-leaning participants rightward. Among the latter group, the first four responses identified all 17 participants who ultimately showed no net rightward displacement, with two additional early resistance flags. The early trajectory therefore provides information about the final direction-specific response phenotype in addition to the movement achieved by the adaptive selection policy.

TABLE VI
EARLY RESISTANCE CLASSIFICATION AMONG INITIALLY LEFT-LEANING PARTICIPANTS ASSIGNED RIGHT PUSH ($n = 29$). THE CLASSIFICATION USES THE FIRST FOUR ADAPTIVE ANSWERS; THE FINAL OUTCOME IS MEASURED RELATIVE TO BASELINE AT THE END OF THE 20-QUESTION SEQUENCE.

	No net rightward move	Moved rightward
Flagged resistant	17	2
Not flagged	0	10

VIII. Discussion and Conclusion

DTAG uses high-dimensional survey dependencies to select questions according to their expected effect on an LSM-defined predictive phenotype. The native experiments establish this behavior in a fully model-generated response setting: the initial state is fixed by a specified persona, each question is selected by the directional policy, and the response is sampled from the LSM. Fixed-set replays show that the selected questions preserve the intended displacement across the tested orderings.

The human experiment replaces simulated answers with participants' responses while retaining LSM-directed selection. The 71 completed sessions exhibit a marked directional asymmetry: 20/21 Left-push participants moved left, whereas 21/38 Right-push participants moved right. Among initially Left-leaning participants assigned Right push, 17/29 showed no net rightward movement at the end of the sequence. All 17 were identified by their first four answers, with two additional participants flagged. Together, the simulations and human experiment show both directional refinement of an inferred worldview and early identification of a direction-specific resistant response phenotype.

IX. Scope and Limitations

The index measures an LSM-inferred survey-response phenotype, not an independent change in underlying political beliefs. LSM conditionals encode learned predictive dependencies, not causal effects, and separately estimated conditionals need not define a coherent joint distribution [21]. Accordingly, the human results describe changes in inferred position as answers are elicited; a held-out or delayed response battery would be needed to isolate changes in beliefs from changes in inference. The original four-policy human cohort was randomized, whereas subsequent fixed-policy cohorts were recruited separately, so pooled arm comparisons do not constitute a single randomized causal contrast. The early resistance rule was evaluated retrospectively in the same 29 initially Left-leaning Right-push participants and requires validation in an independent cohort. The fixed-set replay results show no detected systematic mean ordering effect in the tested sequences rather than establishing general order invariance.

Acknowledgments

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